

if the stock opened at \$29.95, the \$0.05/share move represents a saving to the manager. Depending upon this overnight price movement the manager may change the trading strategy to become more or less aggressive. Even in situations where the portfolio manager uses current market prices in the stock selection process there will likely be some delay (although it may be very short in duration) in determining what stocks and shares need to be purchased and/or sold and releasing those orders to the market. Thus by the time these orders are entered into the market the current market price will be different than the historical price benchmark that was used to determine what stocks to hold in the portfolio.

The historical benchmark price is:

$$E_0[\text{Historical Cost}] = (\bar{P} - P_{hist}) \cdot \text{Side}$$

The E_0 notation is used to denote that this is the expected cost at the beginning of trading. This cost for a buy order and a strategy expressed in terms of trade rate using our formulas above is described as follows:

Let,

$$P_{hist} = \text{Historical Decision Price}$$

$$\bar{P} = P_0 + (b_1 \cdot I' \cdot \alpha) + (1 - b_1) \cdot I'$$

Then the difference is:

$$E_0[\bar{P} - P_{hist}] = (b_1 \cdot I' \cdot \alpha) + (1 - b_1) \cdot I' + (P_0 - P_{hist})$$

If we define the delay cost to be $(P_0 - P_{hist})$ our historical cost is:

$$E_0[\bar{P} - P_{hist}] = MI' + \text{Delay}$$

And we have:

$$E_0[\text{Historical Cost}] = (b_1 \cdot I' \cdot \alpha) + (1 - b_1) \cdot I' + (P_0 - P_{hist}) \quad (8.10)$$

Notice that this is the same cost function as the arrival cost above plus a delay component that is a constant. When a manager selects the previous night's closing price as the benchmark price and starts trading at the open, the "Delay" cost represents the overnight price movement and translates to either a sunk cost or savings to the fund.

Unfortunately, as managers' claim time and time again, this movement represents a sunk cost much more often than it represents a saving because managers as a group do a very good job at figuring out what stocks are mis-priced. Thus, if there is any gap in trading, such as the overnight close, the